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# Diogo Cruz

*Curriculum Vitae*

## General Education

2020-25 **Physics PhD (Quantum Computing)**, *Instituto Superior Técnico (IST)*, Portugal.
Developed quantum algorithms to solve nonlinear partial differential equations; quantum error correction; and tackle problems for near-future quantum devices.

2025 **Visiting Scholar**, *UC Berkeley*, USA.

2024 **Visiting Student**, *MIT*, USA.

2019-20 **Graduate Research Assistant**, *Instituto de Telecomunicações*, Portugal.

2014-19 **Integrated Master Degree (MSc) in Engineering Physics (MEFT)**, *IST*, Lisboa.
Bachelor's and Master's GPA of 18.4 and 18.1/20, resp.; **one of the top students in the course**.

2018 **Erasmus at École Polytechnique Fédérale de Lausanne (EPFL)**, Switzerland.

## AI Safety

2026 **SPAR Spring 2026 - AI Safety Research Program**, *Mentor*.
Mentoring 2 projects, with 8 mentees and expecting 2 papers:

1. Applying the technique of Fluid Benchmarking to efficiently and cheaply run AI safety benchmarks.
2. Continuing previous Algoverse project on operationalizing situational awareness in LLM agents.

2025 **Algoverse AI Safety Fellowship**, *Mentor*.
Mentored 2 projects, each with 3 mentees:

1. How long contexts lead LLM agents to have different safety behaviors, resulted in AAAI 2026 TrustAgent workshop paper, `https://arxiv.org/abs/2512.02445`;
2. Operationalizing situational awareness in LLM agents (ongoing, see above).

2025 **SPAR Fall 2025 - AI Safety Research Program**, *Mentor*.
Mentoring 3 projects, totaling 10 mentees:

1. Developing methods to measure how LLM agents gradually modify their original goals during long-horizon tasks - resulted in 2 papers, each accepted to 2 ICLR workshops (AIWILD & LLA);
2. Investigating how LLM agents behave when their tools become unreliable, and whether this triggers reward hacking - work being extended, planning COLM submission;
3. Testing whether model can suffer from emergent anti-sycophantic behavior - blog post in preparation.

2025 **Introduction to AI Evaluations**, *Invited Speaker*, EAGxSãoPaulo, Brazil.

2025 **CHAI Internship**, Berkeley.
Developed methods to train more robust and performant probes when the amount of labeled samples is limited (as is the case for hard-to-classify or superhuman tasks). Preparing conference paper.

2025 **UK AISI bounty programme**, *Contractor*.
Implemented better scaffolding to study agents solving complex tasks.

2024-25 **Catalyze AI Safety Incubation Program**, *Phases 1 & 2*, London.
Explored safety evals for AI agents, by founding *Anchor Research* (no longer active).

2024-25 **SPAR Spring 2025: Prompt Attacks in Unlearning Methods Project**, *Mentor*.
3 mentees, resulted in COLM SoLaR Workshop paper, `https://arxiv.org/abs/2506.10236`.

2024-25 **AI Safety Camp 10: Multi-turn Jailbreaks Project**, *Research Lead*.
5 mentees, resulted in COLM SoLaR Workshop paper, `https://arxiv.org/abs/2508.07646`.

2024 **Research Engineers Club**, *Safe AI London*.
Replicated WMDP benchmark paper.

2024 **Pivotal Research Fellowship**, *Pivotal*.
Resulted in TMLR paper. `https://arxiv.org/abs/2505.21552`.

2024 **AI Safety Fundamentals: Governance course**, *Participant & Facilitator*, Bluedot Impact.

2023,24 **AI Safety Fundamentals: Alignment course**, *Participant & Facilitator*, Bluedot Impact.

2024 **AI Safety, Ethics, and Society**, *Facilitator*, CAIS.

2023 **ML Safety Scholar Programs**, *CAIS*.

2023 **AI Safety Hub Labs**, *Team Leader*, Oxford.
Neurips SoLaR Workshop paper, `https://arxiv.org/abs/2311.04046`.

2017 **Learning from Data**, *Caltech*, virtual.

2017 **CS231n: Convolutional Neural Networks for Visual Recognition**, *Stanford*, virtual.

## Teaching

2022-23 **Techniques of Mathematical Physics**, *Physics Engineering*, IST.

2022-23 **Quantum Mechanics**, *Aerospace, Naval, Mechanical Engineering*, IST.

2017-18 **Electromagnetism and Optics**, *Aerospace, Naval Engineering*, IST.

## Selected Publications

### AI Safety

2026 **Inherited Goal Drift: Contextual Pressure Can Undermine Agentic Goals**, *Menon, A. et al*, ICLR 2026 AIWILD & LLA Workshops, 2025.

2026 **Asymmetric Goal Drift in Coding Agents Under Value Conflict**, *Saebo, M. et al*, ICLR 2026 AIWILD & LLA Workshops, 2025.

2026 **Unreliable Tools Increase Reward Hacking by AI Coding Agents**, *Hadeliya, T. et al*, In preparation for COLM, 2025.

2025 **Prompt Attacks Reveal Superficial Knowledge Removal in Unlearning Methods**, *Jang, Y. et al*, arXiv: 2506.10236, COLM SoLaR Workshop, 2025.

2025 **Multi-Turn Jailbreaks Are Simpler Than They Seem**, *Yang, X. et al*, arXiv: 2508.07646, COLM SoLaR Workshop, 2025.

2024 **Understanding the learned look-ahead behavior of chess neural networks**, *Cruz, D.*, arXiv: 2505.21552, TMLR, 2024.

2023 **Reinforcement Learning Fine-tuning of Language Models is Biased Towards More Extractable Features**, *Cruz, D. et al*, arXiv: 2311.04046, NeurIPS SoLaR Workshop, 2023.

### Quantum Computing

**See *Google Scholar* for more**.

## Selected Awards

2022 **$3000 1st Prize in Classiq Coding Competition's Hamiltonian Challenge**.

2022 **Winner of Hackathon QCHACK 2022 (QuTech Challenge)**, *Stanford*, United States.

2021 **Winner of Hackathon iQuHACK 2021 (D-Wave Challenge)**, *MIT*, United States.

2019 **Grant from Gulbenkian Program "New Talents in Quantum Technologies"**.

2018 **2nd place for "IBM Q Best Paper Award" (DOI: 10.1002/qute.201900015.)**.

2013-14 **Astronomy Olympiad**.
**1st place** nationally and **Honorable Mention** at the IOAA in 2014 - best result ever for Portugal.

2013-14 **Physics Olympiad**.
**Honorable Mention** at the national level and at the XLV IPhO in 2014, in Kazakhstan.